UNIT 2 · TOPIC 1(C)

Chain Drives and Sprockets

Syllabus topic: Design and selection of machinery elements: gears, pulleys, chains and sprockets, belts

Chains used on farm machinery (Kepner §4.14–4.18)

Chain	Characteristics and use
Precision roller chain	High speed and power (up to ~10 m/s); standard pitches in multiples of 1/4 in.
Detachable-link chain (steel or malleable iron)	Slow, heavily loaded conveyors/elevators, dusty conditions; cheapest; easy to repair in the field.
Pintle chain	Intermediate; barrel-and-pin; tolerates grit (manure spreaders).
Silent (inverted-tooth) chain	Smooth, high speed, costly; rare on farm equipment.

Pitch and chain speed

- Chain pitch = distance between successive pin centres.

$$v = p \cdot T \cdot N/60 \ (\text{m/s, with p in m})$$

Chordal (polygonal) action

- The chain wraps a polygon, not a circle, so its speed varies cyclically with each tooth engagement. The relative variation term is **(1 - cos(180°/T))**.
- The % speed fluctuation drops sharply as tooth number rises and is negligible beyond **~17–19 teeth**. Small sprockets (**<12 teeth**) give rough, noisy drives.
- Use a small sprocket of **≥17 teeth** for high-speed drives. An odd tooth number with even chain pitches equalizes wear.

Installation and standards

- Run chains with slight slack — unlike belts, they need no initial tension. Put the idler/tensioner on the slack side.
- Replace the chain at **~2–3%** elongation.
- Sprocket and roller-chain dimensions are standardized (ASAE/ISO). **Attachment links (A-type, K-type)** carry slats/paddles on conveyors — a uniquely agricultural use.

Practice MCQs

Q15. Chordal (polygonal) action in chain drives causes:

(a) Chain elongation
(b) Cyclic variation in chain velocity
(c) Wear of pins
(d) Sprocket eccentricity

Q16. To minimize chordal action, the smaller sprocket should generally have at least:

(a) 6 teeth
(b) 10 teeth
(c) 17 teeth
(d) 30 teeth

Q17. The chain best suited to gritty, slow-speed farm conveyor service and field repair is:

(a) Silent chain
(b) Precision roller chain
(c) Detachable-link (steel/malleable) chain
(d) Leaf chain

Q18. Chain speed (m/s) with pitch p (m), sprocket teeth T, speed N (rpm) is:

(a) pTN/60
(b) pT/60N
(c) 60pTN
(d) pN/60T

Q19. Unlike belts, chain drives are operated with:

(a) High initial tension
(b) A small amount of slack
(c) Zero backlash
(d) Crowned sprockets

Q20. Attachment links (A- and K-type) on roller chains are used in agriculture to:

(a) Join chain ends
(b) Mount slats/paddles on conveyor/elevator chains
(c) Reduce noise
(d) Increase strength

Answers

Q15 — (b) Cyclic variation in chain velocity

Q16 — (c) 17 teeth — *Chordal velocity variation ∝ (1 - cos 180°/T); below 17 teeth the fluctuation and noise rise steeply.*

Q17 — (c) Detachable-link (steel/malleable) chain

Q18 — (a) pTN/60

Q19 — (b) A small amount of slack

Q20 — (b) Mount slats/paddles on conveyor/elevator chains

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 4 (Elements of Rotary Power-Transmission Systems) and Appendix (ASAE V-belt standards); standard machine design references (Shigley; V.B. Bhandari); ASAE/IS standards.